STUDENT INTRANET

RISE - 503 Capstone Proposal

Student Workspace – Proposal – History **Status:** Proposal - Program Director Approved

Provides immersive, hands-on experience through a capstone project where students solve real-world engineering design challenges in interdisciplinary, often industry-collaborative teams. Emphasizing systems thinking, they integrate subsystems to balance performance, cost, reliability, and sustainability. Students refine project management, communication, and teamwork skills while applying advanced tools and methods in industrial systems and design engineering. Each project culminates in a comprehensive report and formal presentation to faculty, industry partners, and other stakeholders.

Student: O'Keefe, Owen (omokeefe@umich.edu)

Term: FA 2026 **Course:** 503 Capstone

Advisor: Bordley, Robert (rbordley@umich.edu)

Team Project: No

Capstone Coordinator Notes:

Looks like a great project!

Project Information

503 Class: * IOE 503-881

Back Registering (SS semester only): No

Is this project part of a larger team project (up to 4 total)? Only one team member needs to submit a proposal on behalf of the project team: *

No

Sponsor Name: * Owen O'Keefe

Sponsor Email: * owen.okeefe@geaerospace.com

Sponsor Department: * N/A

Project Location: * Grand Rapids, MI

Remote Project? * Yes

Project Type: * Student Initiated Research Topic

Project Type (if "Other"):

Discovery Mode: * Student searching

Discovery Mode (if "Other"):

Expected Project Savings: *

Project provides an intangible value only (no direct cost savings) to the project sponsor

Expected Project Savings (if "Other"):

AI/Automation: Does this project leverage or evaluate AI, Machine Learning, or Workflow Automation? *

Yes

If yes, please briefly state the tools planned:

LLMs for the refinement of requirements and defining system components

NDA needed: * No

503 Capstone Proposal

Project Title: *

An MBSE Architecture of the National Airspace System from Enterprise Objectives to Aircraft Behavior

Project Summary - Overview of the project: *

The project aims to define a comprehensive SysML-based System-of-Systems architectrue of the National Airspace System to serve as a reusable reference model for understanding interactions across organizational and technical boundaries for evaluating how advanced decision-support capabilities can be integrated into the NAS. A representative optimization service, such as trajectory optimization, will be incorporated to demonstrate the practical application of the architecture and its ability to support future operational capabilities.

The NAS is a highly interconnected SoS that is composed of independent organizations, operators, technologies, and physical assets that collectively enable safe, efficient, and equitable use of the airspace. It encompasses regulatory authorities, air navigation service providers, and individual aircraft systems that must coordinate in real time despite differing objectives, authorities, and operational constraints.

Although significant work has been performed to model portions of the NAS, existing architectures frequently focus on either enterprise-level operations, air traffic management, communications infrastructure, or individual aircraft systems in isolation. Relatively little work has been performed to model how strategic objectives and operational constraints propagate through the NAS and ultimately become executable aircraft behavior.

Problem Statement - Identify specific issues that will be addressed with the project: *

The NAS is a complex system in which operational decisions are made by independent actors, each of which may or may not be accounting for the impacts on other system stakeholders, which includes regulatory agencies, air traffic service providers, airport operators, airline operations centers, MRO suppliers, flight crews, passengers, and aircraft systems. While substantial architectural work has been performed within individual domains, the relationships and information flows between these domains are often incompletely represented.

Existing models frequently focus myopically on one actor or problem, exhibiting one or more of the following limitations:
- Emphasis on enterprise-level capabilities without sufficient representation of aircraft behavior and onboard desision-making,
- Emphasis on aircraft systems without adequate representation of the broader operational context within which those systems operate,
- limited traceability between strategic objectives, operational constraints, human decisions, and the resulting aircraft trajectories,
- difficulty evaluating the impacts of new operatioal concepts, automation technologies, and optimization services across organizational boundaries.

As a result, there is no broadly applicable architecture that fully captures how strategic objectives become executable aircraft behavior. This lack of end-to-end traceability limits the ability of engineers, operators, and decision makers to understand the consequences of changes within the NAS and complicates the integration of future decision-support technologies.

This project seeks to address this gap by developing a System-of-Systems architecture that explicitly models the propagation of intent throughout the National Airspace System, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

Project Goals - Expected project outcomes that align with the broader business strategies, larger research goals, or societal impact: *

The primary goal of this project is to develop a reusable and extensible MBSE architecture of the NAS that improves understanding of the relationships among organizations, systems, and operational decisions.

Specific project objectives include:
- Develop a hierarchical SysML representation that captures organizational, operational, informational, and physical domains.
- Identify and model representative elements such as governance actors, airspace management functions, aircraft subsystems and onboard decision systems, and decision-support services such as trajectory optimization, and conflict detection; and
- Establish traceability between:
- Strategic objectives;
- Airspace management decisions;
- Controller actions;
- Flight operations decisions;
- Aircraft system behavior; and
- Vehicle trajectories.
- Define information exchanges and interfaces among stakeholders and systems.

This architecture may improve foresight and communication among stakeholders, facilitate systems engineering analyses, and support evaluation of operational concepts, which may ultimately lead to improved operational efficiency, reduced environmental impact, and more effective utilization of limited airspace resources.

Proposal Approach and Methodologies - All team members' roles must be identified. Explicitly state the connection between the project and student's Program(s): *

The project will employ a Model-Based Systems Engineering (MBSE) methodology using SysML and MagicDraw/Cameo Systems Modeler to develop a multi-level System-of-Systems architecture of the National Airspace System.

The work will proceed through the following phases:
- Phase 1: Stakeholder and Domain ID: identify stakeholders and establish system boundaries based on various criteria (e.g. authority, physical, etc.)
- Phase 2: Architecture Development: Develop structural and behaviro models of the NAS, including (enterprise and sub-level requirements, BDDs, IBDs, etc.)
- Phase 3: Intent to Trajectory Modeling: Develop activity & sequence diagrams relevant to operational actors and traceability, describing through behavioral models how objectives become actions influenced by vehicle & other sub-system dynamics.
- Phase 4: Demonstrate Representative Capability: implement and integrate a representative optimization capability such as trajectory optimization with specific objectives/requirements, using vehicle and system models integrated into the MBSE architecture. One possible method could leverage MOE defined within Cameo model to feed into python-based optimization algorithm with single-objective optimization using a weighted cost function that includes MOEs and leverages vehicle models specified by the Cameo model.

This project directly aligns with the program objectives by applying systems engineering principles to a large-scale, socio-technical System-of-Systems characterized by numerous stakeholders, competing objectives, and complex interactions across organizational and technical boundaries.

The project synthesizes concepts from systems architecture, model-based systems engineering, optimization, decision analysis, and multidisciplinary systems integration. It leverages my professional experience in flight management systems and trajectory optimization to address an important gap in current NAS architectures.

Project Milestone	Deliverable (if applicable)	Lead/Team Member	Expected Completion Date	Target Metric/ Verification Method
PESTLE Analysis & System Definition	Context diagram, stakeholder map	Owen O'Keefe (omokeefe)	2026-09-07	Review
Authority & Responsibility Models	RACII (Responsible, Accountable, Consulted, Informed, Impacted) matrices	Owen O'Keefe (omokeefe)	2026-09-10	All decisions assigned
State Machines	State behavior of key entities	Owen O'Keefe (omokeefe)	2026-09-15	Executable state logic
Sequence Diagrams	Information exchanges	Owen O'Keefe (omokeefe)	2026-09-22	Message completeness
Activity Diagrams	Operational activities	Owen O'Keefe (omokeefe)	2026-10-01	Scenario walkthrough

Project Milestone	Deliverable (if applicable)	Lead/Team Member	Expected Completion Date	Target Metric/ Verification Method
Requirement Traceability	Requirements matrix	Owen O'Keefe (omokeefe)	2026-10-12	Complete allocation
Measures of Effectiveness	MOE/MOP hierarchy	Owen O'Keefe (omokeefe)	2026-10-17	Quantifiable metrics defined
Requirements Definition	Requirements model	Owen O'Keefe (omokeefe)	2026-10-20	Trace to stakeholders
Internal Block Diagrams	IBDs	Owen O'Keefe (omokeefe)	2026-10-25	Critical interfaces connected
Interface Definition	ICD	Owen O'Keefe (omokeefe)	2026-10-29	All interfaces identified
Draft Architecture	BDDs & package structure	Owen O'Keefe (omokeefe)	2026-11-03	All domains represented
Operational Scenarios	ConOps document	Owen O'Keefe (omokeefe)	2026-11-08	1-3 representative scenarios
Systems-of-Interest Definition	Text-based documentation	Owen O'Keefe (omokeefe)	2026-11-13	Clear boundaries & exclusions
Literature Review	Annotated Bibliography	Owen O'Keefe (omokeefe)	2026-11-17	10-20 papers categorized
Optimization Prototype	Working code	Owen O'Keefe (omokeefe)	2026-11-21	Unit test cases
Integration of MBSE Architecture	Traceability links	Owen O'Keefe (omokeefe)	2026-11-24	End-to-end scenario
Final Paper	Thesis/Paper	Owen O'Keefe (omokeefe)	2026-12-18	Revisions
Paper Submittal	Thesis/Paper	Owen O'Keefe (omokeefe)	2026-12-20	Official
Final Baseline & Draft Paper	Versioned model & draft paper	Owen O'Keefe (omokeefe)	2026-12-26	Draft

Safety Risks and Mitigation - Identify safety risks and mitigation strategies associated with your project. These may include: injury to the team, injury to the end user, damage to the equipment, etc.: *

None identified; work is purely digital in nature and will not be integrated into any live system. While my professional work overlaps significantly with the project material, I will not be incorporating any knowledge specific to role and will ensure there is no risk to IP exposure. The intent is to leverage git-hub for version control and cite professional references to ensure information is accurate and acquired from the public domain.

Expected Impact and Value-Add to Sponsor - Identify the value to the sponsor, end users, and/or stakeholders. Consider quantifiable benefits (e.g., financial benefits, time savings, etc.) and intangible benefits (e.g., customer satisfaction, employee morale, safety, etc.). This should emphasize why the project is important to complete: *

The proposed architecture provides a flexible, reusable representation of the NAS that improves understanding of system interactions, reduces the cost and risk of change impact analyses, and enables evaluation of advanced decision-support capabilities and optimization concepts from a holistic, multi-stakeholder perspective. In doing so, it reduces engineering analysis time, ensures a holistic understanding of different sub-system actions & changes, that reduces the cost of future trade studies

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